CMPT 305 Final Project
Group 6: Kenneth Emata, Luis Tanafranca, Cyrus Jabez Rasos

Results of each experiment:

Trace file	depth	starting instruction	time (ms)	%int	%fp	%branch	%load	%store
compute_fp_1	1	1	1.257448	25.7445	28.8089	25.7444	19.7022	0
compute_fp_1	1	5000000	1.257061	25.7059	28.8705	25.7056	19.718	0
compute_fp_1	1	10000000	1.25464	25.4635	29.2584	25.4635	19.8146	0
compute_fp_1	1	15000000	1.032846	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	1	20000000	1.032706	19.4822	25.736	3.2683	38.6455	12.868
compute_fp_1	1	25000000	1.032646	19.46	25.7477	3.2615	38.6569	12.8739
compute_fp_1	2	1	1.19965	25.7445	28.8089	25.7444	19.7022	0
compute_fp_1	2	5000000	1.200098	25.7059	28.8705	25.7056	19.718	0
compute_fp_1	2	10000000	1.202929	25.4635	29.2584	25.4635	19.8146	0
compute_fp_1	2	15000000	0.9411	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	2	20000000	0.941042	19.4822	25.736	3.2683	38.6455	12.868
compute_fp_1	2	25000000	0.941021	19.46	25.7477	3.2615	38.6569	12.8739
compute_fp_1	3	1	0.766459	25.7445	28.8089	25.7444	19.7022	0
compute_fp_1	3	5000000	0.76637	25.7059	28.8705	25.7056	19.718	0
compute_fp_1	3	10000000	0.7658	25.4635	29.2584	25.4635	19.8146	0
compute_fp_1	3	15000000	0.815574	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	3	20000000	0.815662	19.4822	25.736	3.2683	38.6455	12.868
compute_fp_1	3	25000000	0.815715	19.46	25.7477	3.2615	38.6569	12.8739
compute_fp_1	4	1	0.799767	25.7445	28.8089	25.7444	19.7022	0
compute_fp_1	4	5000000	0.800068	25.7059	28.8705	25.7056	19.718	0
compute_fp_1	4	10000000	0.801953	25.4635	29.2584	25.4635	19.8146	0
compute_fp_1	4	15000000	0.770264	19.5213	25.7144	3.2817	38.6254	12.8572
compute_fp_1	4	20000000	0.770348	19.4822	25.736	3.2683	38.6455	12.868
compute_fp_1	4	25000000	0.770398	19.46	25.7477	3.2615	38.6569	12.8739
compute_int_0	1	1	1.033342	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	1	5000000	1.044206	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	1	10000000	1.033343	53.2393	0.8778	3.3184	27.02	15.5445
compute_int_0	1	15000000	1.033342	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	1	20000000	1.033343	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	1	25000000	1.033332	53.244	0.8721	3.3174	27.0199	15.5466
compute_int_0	2	1	0.863813	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	2	5000000	0.879177	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	2	10000000	0.863814	53.2393	0.8778	3.3184	27.02	15.5445
compute_int_0	2	15000000	0.863813	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	2	20000000	0.863814	53.2393	0.8778	3.3184	27.0201	15.5444

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compute_int_0	2	25000000	0.863788	53.244	0.8721	3.3174	27.0199	15.5466
compute_int_0	3	1	0.747452	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	3	5000000	0.741809	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	3	10000000	0.747451	53.2393	0.8778	3.3184	27.02	15.5445
compute_int_0	3	15000000	0.747449	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	3	20000000	0.747452	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	3	25000000	0.747449	53.244	0.8721	3.3174	27.0199	15.5466
compute_int_0	4	1	0.707296	53.2393	0.8778	3.3183	27.0201	15.5445
compute_int_0	4	5000000	0.705978	52.7243	2.8805	4.4473	25.5281	14.4198
compute_int_0	4	10000000	0.707295	53.2393	0.8778	3.3184	27.02	15.5445
compute_int_0	4	15000000	0.707293	53.2397	0.8778	3.3183	27.0196	15.5446
compute_int_0	4	20000000	0.707296	53.2393	0.8778	3.3184	27.0201	15.5444
compute_int_0	4	25000000	0.707284	53.244	0.8721	3.3174	27.0199	15.5466
srv_0	1	1	1.19892	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	1	5000000	1.194171	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	1	10000000	1.196871	47.7866	0.5066	18.488	22.8909	10.3279
srv_0	1	15000000	1.196955	47.6417	0.6309	18.4237	22.928	10.3757
srv_0	1	20000000	1.199084	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	1	25000000	1.199455	47.5057	0.4098	18.7701	22.967	10.3474
srv_0	2	1	1.000401	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	2	5000000	0.997132	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	2	10000000	0.998868	47.7866	0.5066	18.488	22.8909	10.3279
srv_0	2	15000000	0.999241	47.6417	0.6309	18.4237	22.928	10.3757
srv_0	2	20000000	1.000373	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	2	25000000	1.000687	47.5057	0.4098	18.7701	22.967	10.3474
srv_0	3	1	0.754912	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	3	5000000	0.753103	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	3	10000000	0.754099	47.7866	0.5066	18.488	22.8909	10.3279
srv_0	3	15000000	0.754247	47.6417	0.6309	18.4237	22.928	10.3757
srv_0	3	20000000	0.75475	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	3	25000000	0.755216	47.5057	0.4098	18.7701	22.967	10.3474
srv_0	4	1	0.713361	47.5764	0.4487	18.7246	22.9315	10.3188
srv_0	4	5000000	0.711852	47.9605	0.7024	18.2051	22.8653	10.2667
srv_0	4	10000000	0.712644	47.7866	0.5066	18.488	22.8909	10.3279
srv_0	4	15000000	0.7129	47.6417	0.6309	18.4237	22.928	10.3757
srv_0	4	20000000	0.713179	47.5779	0.4046	18.7975	22.9112	10.3088
srv_0	4	25000000	0.713632	47.5057	0.4098	18.7701	22.967	10.3474

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(in ms)

Trace File	1	2	3	4	Row Sum	Row Mean	Row Effect
compute_fp_1	1.1	1.07	0.79	0.79	3.79	0.95	0.048
compute_int_0	1.04	0.8	0.75	0.71	3.37	0.84	-0.062
srv_0	1.20	1.00	0.75	0.71	3.66	0.92	0.018
Column Sum	3.38	2.9	2.29	2.21	10.82		
Column Mean	1.1	0.98	0.76	0.74		0.9025	
Column Effect	0.225	0.078	-0.139	-0.165			

Impact of Each Level: Deeper pipelines are faster: D=1 is the slowest (+0.225 ms from the mean) and D=4 is the fastest (−0.164 ms). For traces, compute_int_0 is fastest (−0.063 ms), srv_0 is near average (+0.015 ms), and compute_fp_1 is slowest (+0.048 ms).

Allocation of variance: Pipeline depth accounts for 81.40% of the total variation in execution time, the trace workload accounts for 6.65%, their interaction accounts for 3.95%, and the remaining 7.99% is due to experimental error.

Member Contribution:
Kenneth Emata: Setup Work Environment, Setup Code Boilerplate, Implement Headers and Main function, Format Report
Luis Tanafranca: Implemented Pipeline Simulation Logic, Hazard Handling (Data/Control/Structural), and Performance Metrics Output
Cyrus Jabez Rasos: Debugging and error handling, code testing, results tables, and two-factor analysis calculations